Credit Risk Portfolio Analysis

PD Model Development, Validation & Active Loan Book Scoring

Prepared 2026-09-16 | Credit Risk project

1. Executive Summary

This report documents a probability-of-default (PD) model built on a LendingClub-style loan portfolio and applies it to score the currently active loan book. The training population consists of 87,891 completed loans (issued 2015) with a known outcome — 20.0% charged off overall. A gradient boosting model (HistGradientBoosting) was trained on origination-time features only and validated on a 25% held-out test set, achieving an AUC-ROC of 0.744 and a KS statistic of 0.360 — strong separation between loans that went on to default and those that were repaid.

Applying the model to the 11,402 loans currently outstanding (status "Current") produces a portfolio-level average PD of 33.0% against total exposure of $228.5M. Under a conservative assumption of 100% loss given default (no recovery data is available in this dataset), the resulting expected-loss upper bound is $74.1M. The active book's average PD is materially higher than the 20.0% historical base rate; this is a portfolio-composition effect, not a model calibration problem — the current book is more heavily weighted toward higher-risk grades (C/D/E) than the training population (see Section 5).

Three deliverables accompany this report: an Excel workbook with the full scored loan book, live pivot-style summary tables and charts; an interactive web dashboard for ongoing monitoring; and this document, covering methodology, findings, and recommendations.

2. Data & Methodology

2.1 Data sources

- Training set: 87,891 completed loans (loan_status = Fully Paid or Charged Off), issued in 2015. Target variable: default = 1 if Charged Off, else 0.
- Scoring set: 11,402 currently active loans (loan_status = Current) to be scored for risk.
- Both files were pre-cleaned (no missing values): a co-borrower field block was dropped, employment length and credit history were converted to numeric features, and remaining nulls were imputed using column-appropriate rules (median, zero, or -1 sentinel depending on field semantics).

2.2 Leakage control

A PD model must only use information available at the time of loan origination — using fields that are only observed later (during or after the loan's life) inflates validation metrics but is not usable in production. The following fields were identified as post-origination and excluded from the feature set:

- total_rec_prncp, total_rec_int, total_rec_late_fee — payments received over the life of the loan.
- last_fico_range_high / last_fico_range_low — the borrower's most recently reported FICO score, updated periodically as the loan matures.

The remaining ~191 features (after one-hot encoding categoricals) are all known at, or derivable from, the loan application: loan terms (amount, rate, term, installment), credit bureau attributes (FICO at origination,

DTI, revolving utilization, trade line counts and ages, delinquency history), and categorical fields (grade, sub-grade, purpose, home ownership, state, verification status, application type).

2.3 Model & validation

Two models were trained and compared: a logistic regression baseline (AUC 0.742) and a HistGradientBoostingClassifier (AUC 0.744). The gradient boosting model was selected as it outperformed the baseline while remaining well-calibrated (Brier score 0.139). All metrics are reported on a stratified 25% held-out test set (21,973 loans) not used in training.

3. Exploratory Findings

Default rates vary sharply and monotonically with LendingClub's own assigned grade, confirming that grade — and the interest rate it drives — already captures most of the risk signal available at origination:

Grade	Historical default rate	Loan count
A	5.3%	17,059
B	13.2%	28,027
C	23.7%	24,487
D	34.5%	10,951
E	43.8%	5,498
F	52.8%	1,532
G	59.1%	337

Table 1. Actual charge-off rate by grade, training set.

By loan purpose, small business and house loans carry the highest historical default rates, while car loans and credit-card consolidation carry the lowest:

Purpose	Default rate	Loan count
Small Business	30.0%	779
House	26.8%	280
Moving	22.7%	534
Medical	22.3%	1,037
Debt Consolidation	21.8%	49,644
Renewable Energy	20.4%	54
Other	19.3%	4,908
Major Purchase	19.2%	1,765
Home Improvement	17.9%	5,586
Vacation	17.8%	489
Credit Card	16.4%	22,030
Car	15.0%	785

Table 2. Actual charge-off rate by loan purpose, training set.

60-month loans default far more often than 36-month loans (37.4% vs 14.9%), and interest rate is the single strongest numeric correlate of default (r = 0.31), followed by FICO (r = -0.13) and DTI (r = 0.11).

4. PD Model Performance

4.1 Discrimination (decile / lift table)

Sorting the test set into ten equal-sized groups by predicted PD shows the model cleanly separates risk: the highest-risk decile has an actual default rate of 51.1%, roughly 17x the lowest-risk decile's 2.9%.

Decile (10=highest risk)	Loans	Actual defaults	Avg predicted PD	Actual default rate
10	2,198	1,124	52.5%	51.1%
9	2,197	775	36.0%	35.3%
8	2,197	627	27.4%	28.5%
7	2,197	495	21.8%	22.5%
6	2,197	403	17.9%	18.3%
5	2,198	332	14.5%	15.1%
4	2,197	248	11.5%	11.3%
3	2,197	196	8.7%	8.9%
2	2,197	137	6.1%	6.2%
1	2,198	64	3.5%	2.9%

Table 3. Decile lift table, held-out test set.

4.2 Feature importance

Permutation importance (drop in test AUC when a feature is shuffled) confirms interest rate — itself LendingClub's own risk pricing — dominates, followed by loan term and a handful of credit-utilization and account-activity features:

Rank	Feature	Importance (ΔAUC)
1	int_rate	0.1094
2	term_36 months	0.0236
3	acc_open_past_24mths	0.0057
4	installment	0.0029
5	dti	0.0027
6	loan_amnt	0.0025
7	grade_B	0.0021
8	emp_length_num	0.0019
9	num_rev_tl_bal_gt_0	0.0012
10	credit_history_years	0.0012
11	home_ownership_MORTGAGE	0.0010
12	avg_cur_bal	0.0010

Table 4. Top permutation-importance features, HistGradientBoostingClassifier.

5. Current Portfolio Risk Assessment

The PD model was applied to all 11,402 currently active loans. Risk bands were defined judgmentally on predicted PD to roughly track historical grade-level default rates:

Risk band	Loan count	% of book	Exposure
Low	157	1.4%	$3.0M
Low-Medium	844	7.4%	$17.3M
Medium	1,688	14.8%	$35.5M
Medium-High	2,497	21.9%	$52.3M
High	3,429	30.1%	$68.0M
Very High	2,787	24.4%	$52.4M

Table 5. Active loan book by risk band.

$68.0M + $52.4M of exposure sits in the two highest risk bands combined. Cross-tabulating against LendingClub's own grade shows the model recovers the expected ordering (grade A/B loans cluster in Low/Low-Medium bands; F/G loans cluster in High/Very High), which is a useful sanity check on the model rather than a new finding.

Table 6 lists the 15 highest-risk active loans by predicted PD — the recommended starting point for portfolio review or early-intervention outreach.

Loan ID	Grade	Amount	Purpose	Predicted PD
65875967	F5	$16.5K	debt consolidation	81.6%
65607978	G3	$11.1K	small business	79.6%
67325233	E3	$24.0K	debt consolidation	79.4%
66423766	D3	$18.0K	debt consolidation	77.4%
68062119	E2	$14.0K	house	77.4%
65595586	E1	$14.0K	debt consolidation	76.9%
64711889	E4	$21.8K	credit card	76.8%
64400205	E4	$13.1K	credit card	76.7%
63290632	F3	$24.2K	debt consolidation	76.7%
68574887	E2	$25.0K	small business	76.3%
63844524	F4	$24.7K	small business	76.3%
65412129	E5	$14.4K	debt consolidation	76.3%
63485385	F3	$10.8K	debt consolidation	76.2%
68603483	D4	$25.0K	debt consolidation	76.0%
66561734	D2	$20.0K	credit card	75.5%

Table 6. Top 15 highest-risk active loans. Full list of 20 in the accompanying Excel workbook and dashboard.

6. Recommendations

- Prioritize review of the 2,787 "Very High" risk-band loans ($52.4M exposure) for early intervention — proactive outreach, restructuring, or provisioning.
- Treat the elevated portfolio-average PD (33.0% vs 20.0% historical base rate) as a composition signal: recent originations skew toward C/D/E grades, so overall portfolio provisioning should be reviewed against this shift rather than assumed to track historical averages.
- Because expected loss here assumes 100% loss given default, treat the $74.1M figure as an upper bound. Incorporating actual recovery/collateral data (not present in this dataset) would materially reduce it and should be a near-term follow-up.
- Revisit risk-band thresholds periodically against realized outcomes as active loans season and resolve, rather than treating the current PD-to-band mapping as fixed.

7. Limitations & Assumptions

- Training data covers loans issued in a single year (2015); the model has not been validated against other vintages or macroeconomic regimes, so its accuracy under different economic conditions is unknown.
- Expected loss uses LGD = 100% in the absence of recovery data — a deliberately conservative upper bound, not a calibrated loss estimate.
- Risk-band cut points (Section 5) were set judgmentally, not derived from a formal loss-optimization exercise.
- The model has not been assessed for disparate impact across protected classes; before any operational or pricing use, a fair-lending review is recommended.
- This analysis is for internal portfolio-monitoring purposes and does not constitute investment, credit, or legal advice.